

UC MATH 600 MATHEMATICS FOR COMPUTER SCIENCE · MODULE 11 · LECTURE NOTES

Discrete Probability

Counting with weights, and the errors everyone makes.

MODULE OUTCOMES

- Set up a probability space correctly.
- Use conditional probability and independence.
- Apply Bayes' rule and explain base rates.
- Avoid the standard probability errors.
- Connect this to the randomised algorithms ahead.

1 The setup

NAME THE SAMPLE SPACE BEFORE COMPUTING ANYTHING, BECAUSE ALMOST EVERY ERROR IS A SAMPLE SPACE ERROR

A probability space is a set of outcomes together with a weight on each, summing to one — and an event is simply a subset of the outcomes, whose probability is the sum of the weights in it. Nothing more elaborate is going on.

Which means a probability question is a counting question with weights (Module 06) — and the sum rule, product rule, bijection rule, and inclusion-exclusion all carry over directly, which is why that module comes before this one.

And the classic paradoxes are almost all disputes about the sample space rather than about the arithmetic — 'I have two children and one is a boy' has different answers depending on how you came to learn that, because the set of outcomes consistent with your information differs.

So write the outcomes down, explicitly, especially when the answer surprises you — which resolves more confusion than any formula, and is this module's one habit if you take nothing else from it.

2 Conditioning

$$P(A \mid B) = P(A \cap B) / P(B)$$

- Restrict the sample space to B and renormalise so the weights sum to one again — that is literally all conditioning is, and seeing it that way removes most of the mystery from Bayes' rule in §3.

- **A and B are independent exactly when $P(A \cap B) = P(A) \cdot P(B)$, equivalently when $P(A|B) = P(A)$:** knowing B happened tells you nothing about A. **This is a property to be checked, not an assumption to be made**, and **assuming it is the second commonest error** in the subject (§4).
- **The union bound says $P(A \cup B) \leq P(A) + P(B)$, always — no independence required, no conditions at all** — and it extends to any number of events by induction.
- **The union bound needs no assumptions and is usually enough, which makes it the most useful inequality in CSCE 658:** it is crude, it is free, and the typical randomised-algorithm argument ('the probability that any of n bad things happens is at most n times the probability of one') is exactly this.

3 Bayes and base rates

SETUP

1 person in 10,000 has the condition
the test is 99% accurate in both directions
you test positive. What is
 $P(\text{condition} \mid \text{positive})$?

COUNT 1,000,000 PEOPLE

100 have it. 99 of those test positive.
999,900 do not have it. 1% of them -- that is
9,999 people -- test positive anyway.

total positives: $99 + 9,999 = 10,098$
true positives among them: 99

$P(\text{condition} \mid \text{positive}) = 99 / 10,098$
= about 1%

SO A 99% ACCURATE TEST GIVES A 1% ANSWER.

Not a trick: the false positives are drawn from a
pool 10,000 times larger, so they swamp the true
positives completely.

AND THE LESSON: the base rate is not a detail.

Count a million people instead of applying the formula — the frequency framing makes this obvious where the algebra makes it opaque, which is a measured effect and not merely a preference (CSCE 679 Module 09 §3 covers the evidence). **This is the single most practically important calculation in the module**, and it is worth being able to produce from scratch rather than recalling the conclusion.

WHICH GENERALISES: A RARE THING DETECTED BY AN IMPERFECT TEST IS MOSTLY FALSE POSITIVES

Intrusion detection, spam filtering, disease screening, and fraud detection all have exactly this shape — and all of them produce alert fatigue for precisely this reason (CSCE 701 Module 09's detection material, where the base rate is the central design constraint).

And it is why precision and recall are reported separately rather than collapsed into a single accuracy figure (CSCE 633's evaluation material, CSCE 670 Module 03) — accuracy on an imbalanced problem is dominated by the majority class and says almost nothing.

A classifier that is 99% accurate on a one-in-ten-thousand problem can be worse than useless — it can be beaten by always answering 'no' — and 'accuracy' is exactly the metric that hides this (CSCE 676 Module 13's proxy problem).

So the habit to build: when somebody tells you an accuracy, ask for the base rate — which is one question and changes the interpretation entirely, and is answerable in every honest case.

4 The standard errors

- **Ignoring the base rate (§3) — the most consequential of the five, and the one trained professionals make** in medicine and in security alike.
- **Assuming independence — which is a claim about the world and is usually false for the things you actually care about: correlated failures are the reason redundancy disappoints**, since the two replicas share a power supply, a rack, a library version, or a deployment (CSCE 678's correlated-failure material).
- **Confusing $P(A|B)$ with $P(B|A)$ — 'most accidents happen near home' and 'driving near home is dangerous' are different claims, and the first is mostly a statement about where people drive.** This is the prosecutor's fallacy, and it has sent people to prison.
- **The gambler's fallacy** — independent trials have no memory, and **a coin that has come up heads five times is not 'due'**. The related error in the other direction is assuming a streak means the coin is biased.
- **And conditioning on the wrong thing — selection effects, where the sample you can observe was filtered by something related to what you are measuring** (CSCE 676 Module 08's survivorship, and CSCE 632 Module 02 §2's analytics argument).

AND WHERE THIS GOES NEXT

CSCE 658 is this module plus Module 12's concentration bounds — **randomised algorithms are probability applied to a sample space you constructed yourself, which is the easy case**: you know the distribution exactly because you chose it.

And every machine learning course is this module with continuous distributions — the discrete version is where the intuitions get built, and continuous probability adds machinery rather than ideas.

Plus the probabilistic method: prove that an object with some property exists by showing a randomly chosen object has it with positive probability — which is a counting argument in disguise (Module 06) and is one of the most elegant techniques in combinatorics.

So: name the space, check independence, and ask for the base rate — three habits, and they cover most of what goes wrong in this subject for the rest of your career.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lehman, Leighton & Meyer, chapters 17 to 19 (free)	The whole module , free — and the sample-space discipline of §1 is emphasised throughout, correctly.
Mitzenmacher & Upfal, chapters 1 to 3	§§1 to 3 — probability for computer scientists, and the direct prerequisite for CSCE 658. Library copy.
Gigerenzer — Reckoning with Risk	§3 — the frequency framing and why it works, with the medical examples. Library copy.

Resource	What it covers for this module
Tijms — Understanding Probability	§§1 and 4 — the paradoxes resolved by naming the sample space, which is the right way to meet them. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write the sample space** for five problems before computing.
- 2 **Resolve the two-children problem** both ways, with the spaces written out.
- 3 **Compute five conditional probabilities** from the definition.
- 4 **Check independence** for four event pairs, and find one that is not.
- 5 **Apply the union bound** to a problem with n bad events.
- 6 **Do the 99% test calculation** by counting a million people.
- 7 **Redo it** with base rates of 1 in 100 and 1 in 1,000,000.
- 8 **Find a published accuracy claim** and ask for its base rate.
- 9 **Find an example of each of the five errors** in the wild.
- 10 **Prove something exists** by the probabilistic method.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What is a probability space, and what is an event?
- 2 Why is naming the sample space the key habit?
- 3 Define conditional probability and say what it does.
- 4 Define independence, and say why it is a claim.
- 5 State the union bound and say what it assumes.
- 6 Work the 99% test problem by counting.
- 7 Why does a 99% accurate test give a 1% answer?
- 8 Where does this shape reappear in the program?
- 9 Name the five standard errors.
- 10 What are the three habits this module asks for?